The code is saved in a .ipynb file. This ReadMe assumes the code is being run locally on an IDE.

Prerequisites

Install the required Python packages:
Required Code:
pip install numpy pandas scikit-learn scipy matplotlib seaborn umap-learn hdbscan

Dataset

Place the dataset file “star-galaxy-quasar.csv” in the same directory as the notebook before running.

Running the notebook

1. Open IT1244_Team26_Code.ipynb in your local IDE (for e.g., VSCode)
2. Run all cells in order from top to bottom.
3. The notebook will preprocess the data, apply PCA and UMAP, run K-Means, GMM, Agglomerative Clustering, and HDBSCAN, evaluate models using conductance scores, and print the final ARI score.
4. On completion, a “model.pkl” file will be saved in the same directory containing the final model, original data, and UMAP embeddings.

The file "model.pkl" contains the final model together with the relevant data (both untransformed and transformed under UMAP) associated with the model, in a dictionary object with the following items: model_with_data = {
    'model': final_model,
    'original_data': X_raw,
    'transformed_data': embedding
}

To load the model and data from the .pkl file, run the following:

```
with open('model.pkl', 'rb') as f:
  d = pickle.load(f)
data = d['original_data']
data_transformed = d['transformed_data']
model = d['model']
labels = model.labels_
```